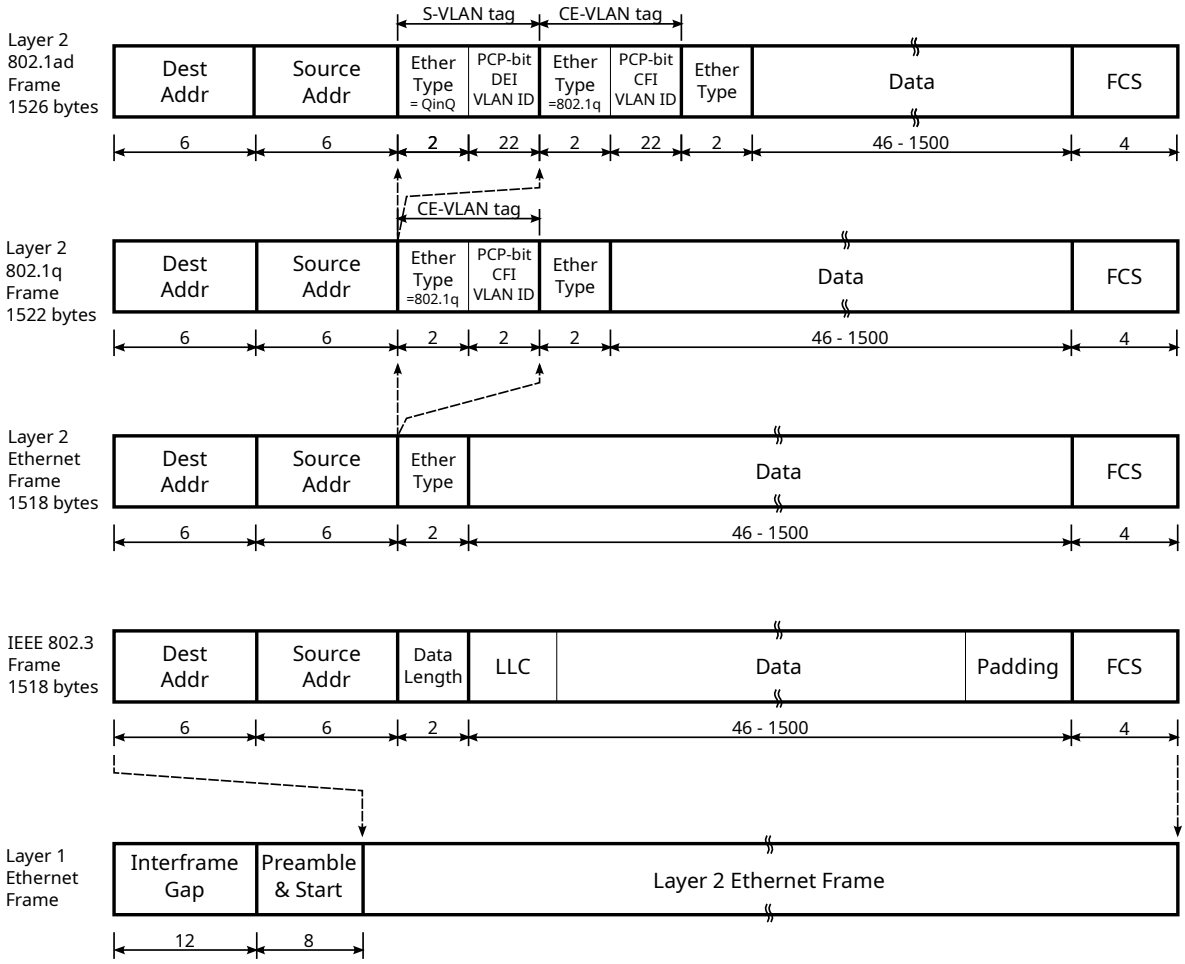
Datos para las evaluaciones

Redes de Comunicaciones - TB067

Comunicación de Datos - 86.12

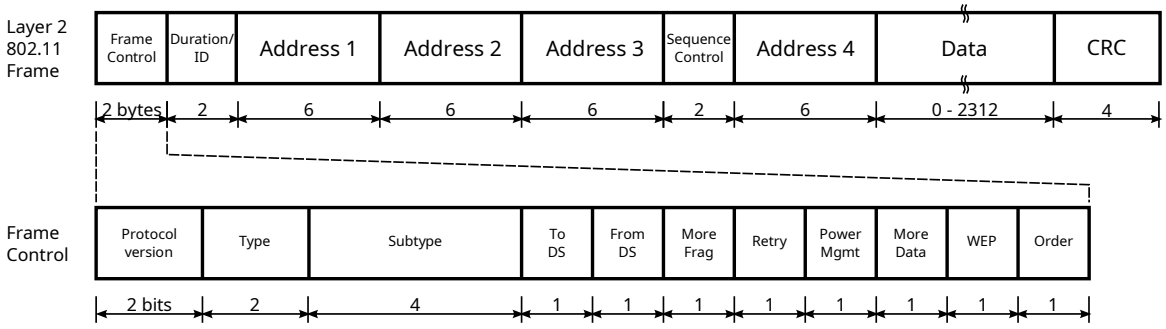
Formato de la trama Ethernet y 802.3



Ethernet Type

0x0800	Internet Protocol Version 4 (IPv4)
0x0806	Address Resolution Protocol (ARP)
0x86DD	Internet Protocol Version 6 (IPv6)
0x8100	802.1q - Customer VLAN Tag Type
0x88A8	802.1ad - Service VLAN tag identier

Formato de la trama MAC 802.11



Direcciones en la trama MAC 802.11

To DS (AP)	From DS (AP)	Address 1	Address 2	Address 3	Address 4	Operación
0	0	DA	SA	BSSID	-	Trama de una STA a otra en un IBSS o Ad Hoc
0	1	DA	BSSID	SA	-	Trama del AP hacia una STA
1	0	BSSID	SA	DA	-	Trama de una STA hacia el AP
1	1	RA	TA	DA	SA	Trama intercambiada entre dos APs

RA: Receiver Address
TA: Transmitter Address
SA: Source Address
DA: Destination Address
BSSID: Basic Service Set ID (Para modo Infraestructura BSS es la dirección MAC del AP, para modo Ad Hoc o IBSS suele tener un valor aleatorio)

Frame Types and Subtypes

Type Value (Bit 3, Bit 2)	Type Description	Subtype Value (Bit 7, Bit 6, Bit 5, Bit 4)	Subtype Description
00	Management	0000	Association request
00	Management	0001	Association response
00	Management	0010	Reassociation request
00	Management	0011	Reassociation response
00	Management	0100	Probe request
00	Management	0101	Probe response
00	Management	0110–0111	Reserved
00	Management	1000	Beacon
00	Management	1001	Announcement traffic indication frame (ATIM)
00	Management	1010	Disassociation
00	Management	1011	Authentication
00	Management	1100	Deauthentication
00	Management	1101–1111	Reserved
01	Control	0000–1001	Reserved
01	Control	1010	PS-Poll
01	Control	1011	RTS
01	Control	1100	CTS
01	Control	1101	Acknowledgment (ACK)
01	Control	1110	CF-End
01	Control	1111	CF-End+CF-Ack
10	Data	0000	Data
10	Data	0001	Data+CF-Ack
10	Data	0010	Data+CF-Poll
10	Data	0011	Data+CF-Ack+CF-Poll
10	Data	0100	Null function (no data)
10	Data	0101	CF-Ack (no data)
10	Data	0110	CF-Poll (no data)
10	Data	0111	CF-Ack+CF-Poll (no data)
10	Data	1000–1111	Reserved
11	Reserved	0000–1111	Reserved

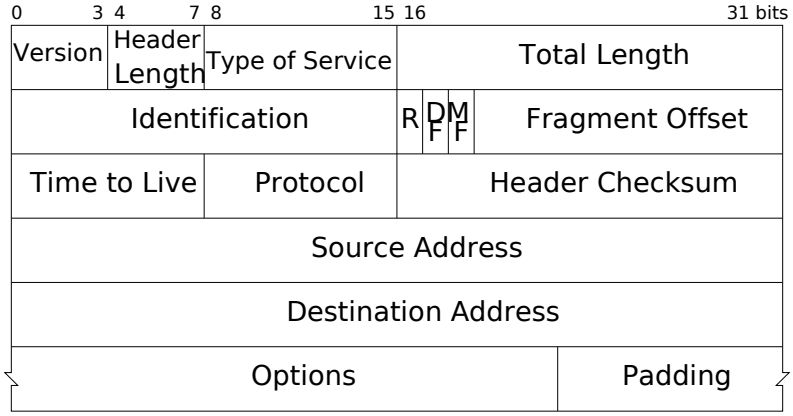
Formato del paquete ARP

Hard Type	Prot Type	Hard Size	Prot Size	Opcode	Sender H Addr	Sender P Addr	Target H Addr	Target P Addr
<2B>	<2B>	<1B>	<1B>	<2B>				

Opcode

1	ARP request
2	ARP reply

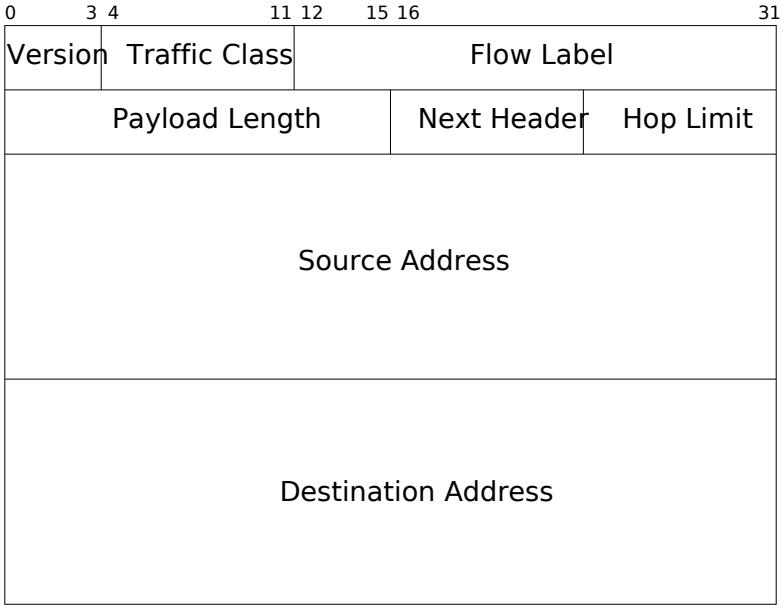
Formato del encabezado IP



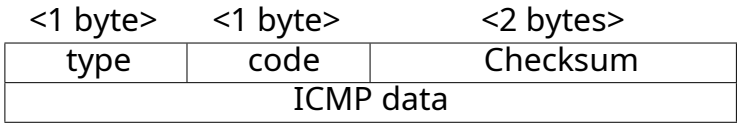
Protocol

1	ICMP (Internet Control Message Protocol)
6	TCP (Transmission Control Protocol)
17	UDP (User Datagram Protocol)

Formato del encabezado IP versión 6



ICMP



Type Name
---- ----------
0 Echo Reply

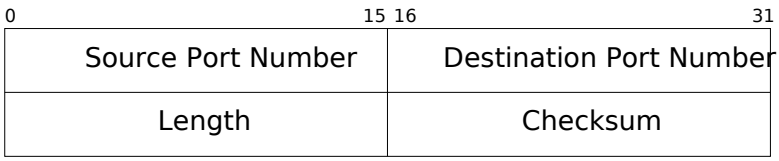
3 Destination Unreachable
Codes
- 0 Net Unreachable
- 1 Host Unreachable
- 2 Protocol Unreachable
- 3 Port Unreachable
- 4 Fragmentation Needed and Don't Fragment was Set
- 6 Destination Network Unknown
- 7 Destination Host Unknown

5 Redirect
Codes
- 0 Redirect Datagram for the Network (or subnet)
- 1 Redirect Datagram for the Host
- 2 Redirect Datagram for the Type of Service and Network
- 3 Redirect Datagram for the Type of Service and Host

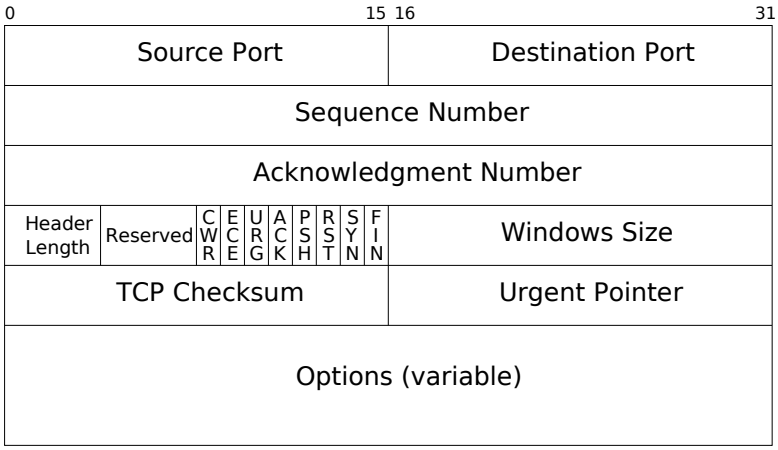
8 Echo

11 Time Exceeded
Codes
- 0 Time to Live exceeded in Transit
- 1 Fragment Reassembly Time Exceeded

Formato del encabezado UDP



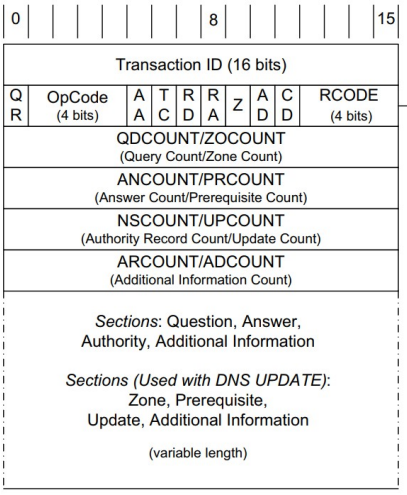
Formato del encabezado TCP



Teoría de colas

$$\rho = \lambda/\mu \qquad T = \frac{1}{\mu - \lambda} \qquad N = \lambda T$$

DNS



Flags:
QR: Query(0)/Response(1)
AA: Authoritative Answer
TC: Truncated Answer
RD: Recursion Desired
RA: Recursion Available
Z: Zero
AD: Authentic Data [RFC4035]
CD: Checking Disabled [RFC4035]

OpCodes (common values):
Query (0) – Regular Query
Notify (4) – DNS NOTIFY [RFC1996]
Update (5) – DNS UPDATE [RFC2136]

RCODEs (common values):
NoError (0) – No Error
FormErr (1) – Format Error
ServFail (2) – Server Failure
NXDomain (3) – Non-existent Domain
NotImp (4) – Not Implemented
Refused (5) – Query Refused

Canal

$$\textbf{Nyquist: } C[bps] = 2BR_1 \qquad R_1\text{: relación bps/baudios}$$

$$\textbf{Shannon: } C[bps] = B\log_2(1 + \frac{S}{N})$$

$$\frac{E_b}{N_0} = \frac{S}{kTR} \qquad N_0 = kT$$

donde $k = 1{,}38 \times 10^{-23}\ J/K$